Melanoma Heterogeneity and Tumor Microenvironment Interactions via Single-Cell RNA Sequencing

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The Dataset

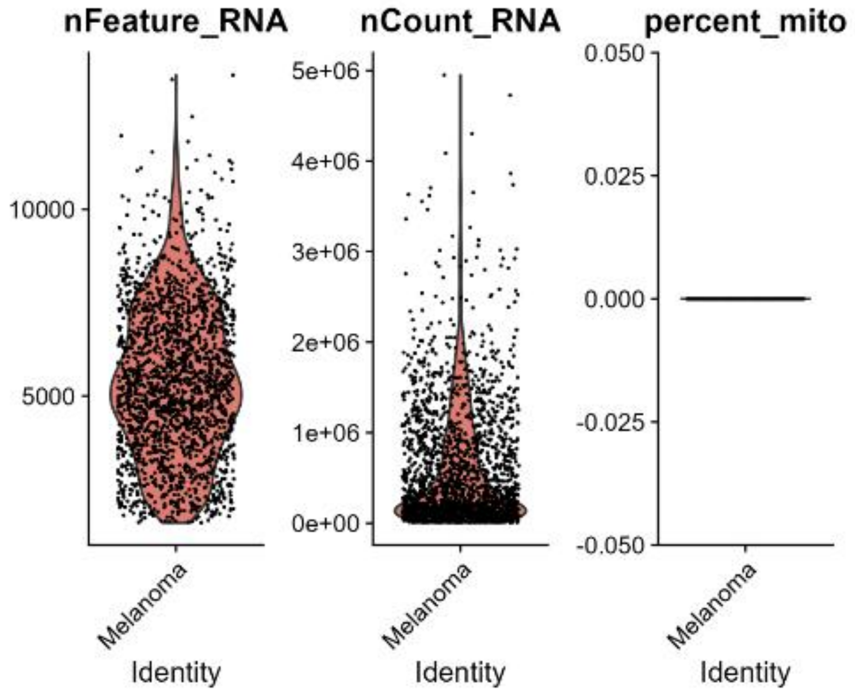
scRNA-seq combined from two melanoma studies of patients under different therapies.

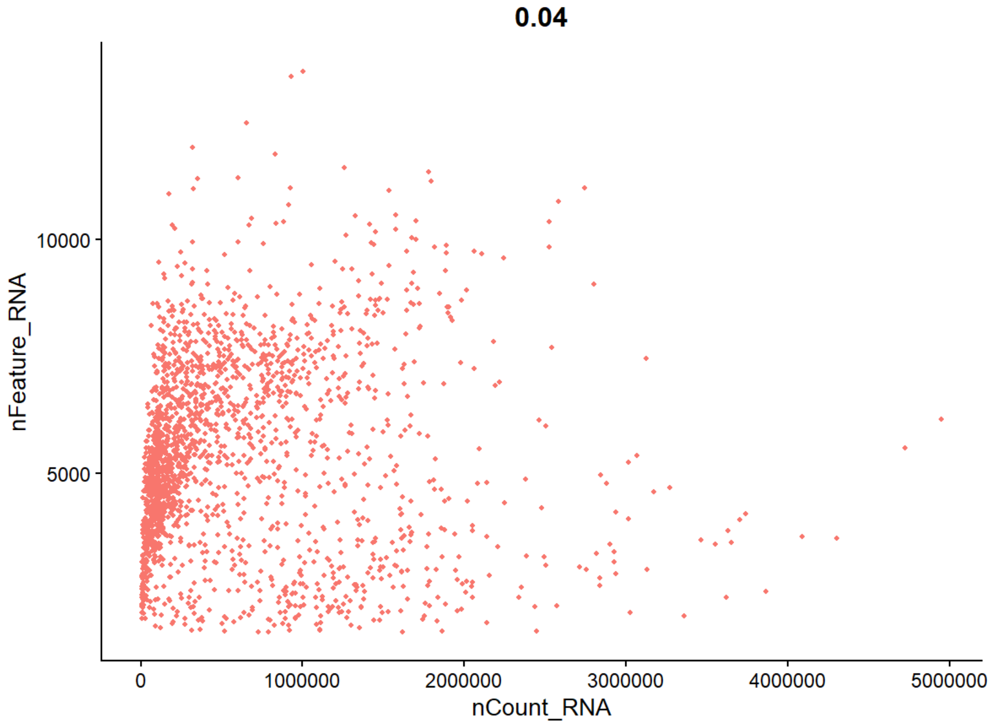
Contains **both tumour and non-tumour cells**.

Provided in **10x format**: gene-cell matrix, gene IDs, cell barcodes, and metadata.

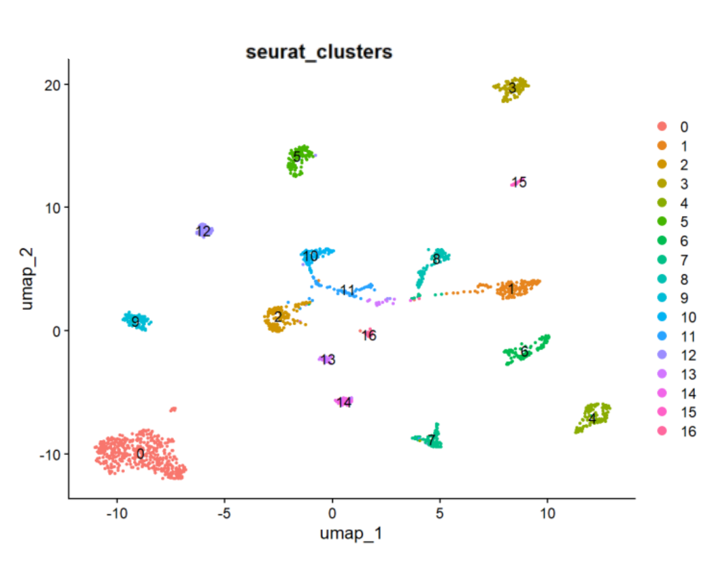
Full data set was used in this analysis

Quality Control

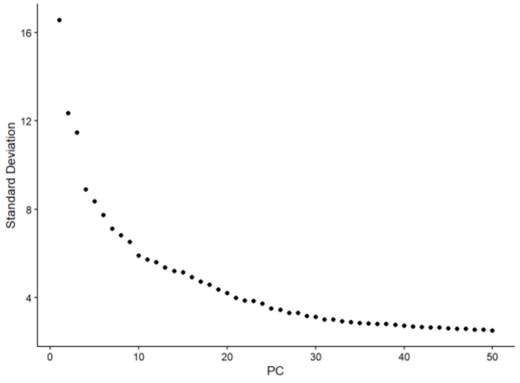




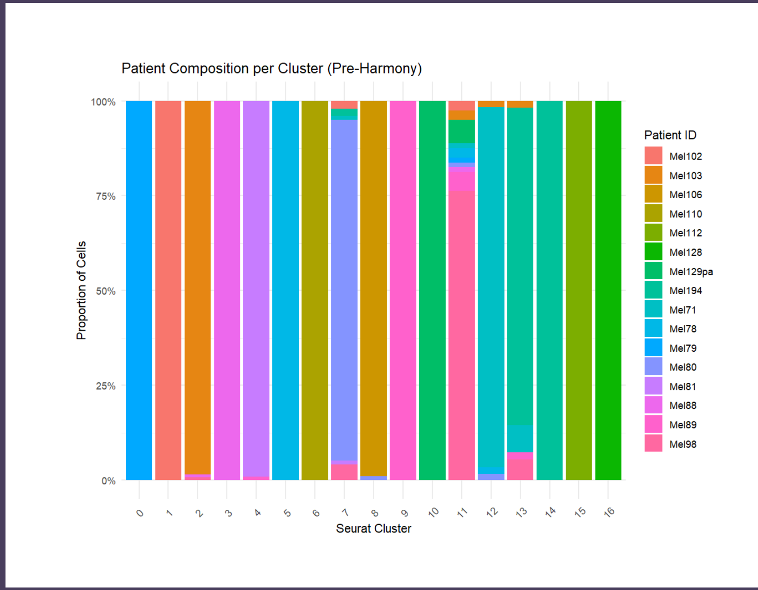
- To remove low quality cell and outliers
- No mitochondrial content detected in this data set
- Filtered to retain cells with 1,000 -10,000 genes

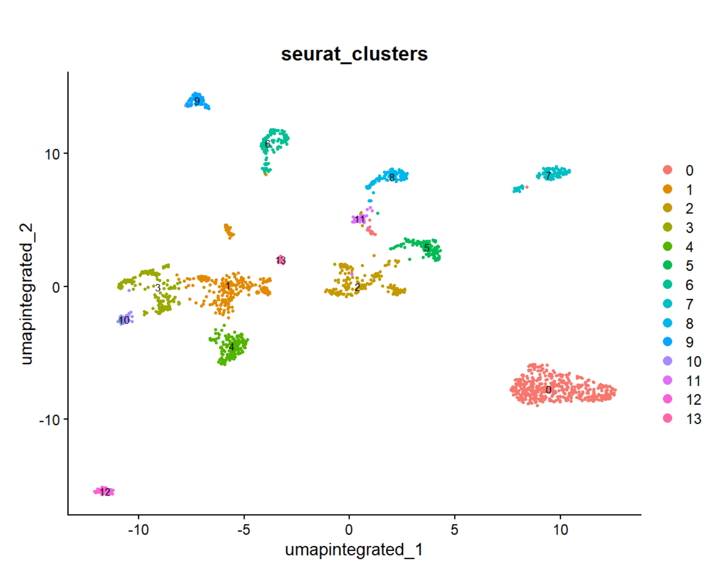


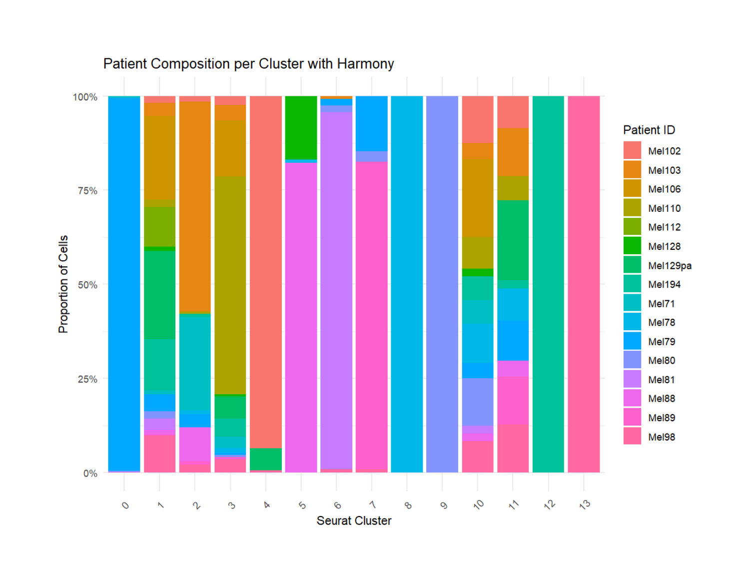
Dimensionality Reduction and Integration



- Clusters initially dominated by individual samples -strong batch effects.
- After integration, cells from different patients mix within clusters.
- Some clusters remain patient-enriched, potential **inter-tumor heterogeneity.**

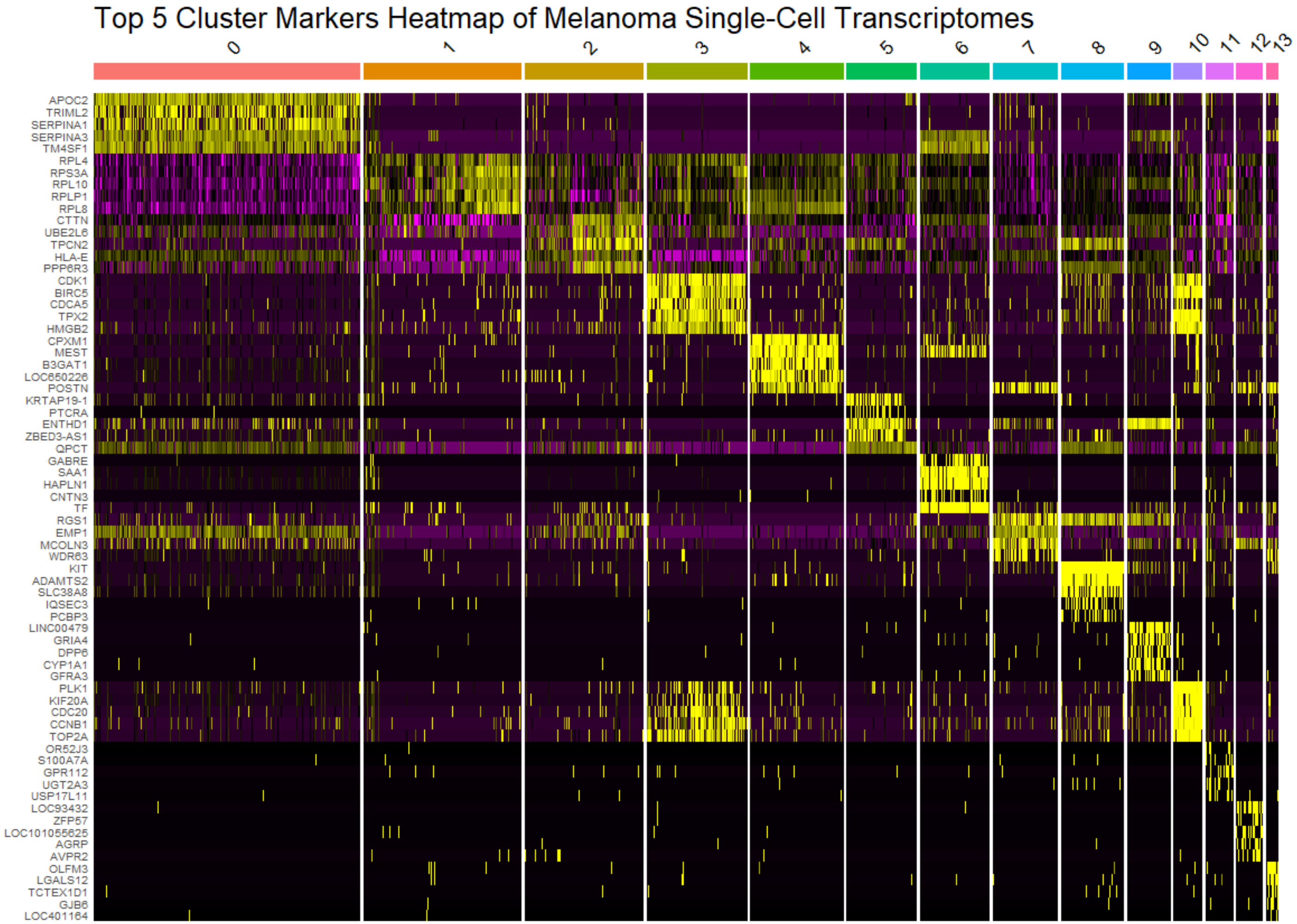






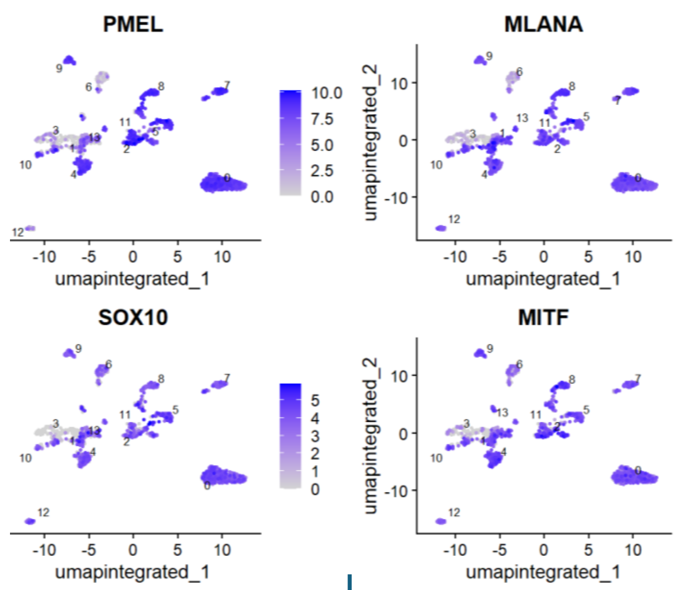
Clusters have unique “genetic fingerprints” distinguishing them 14 identified clusters

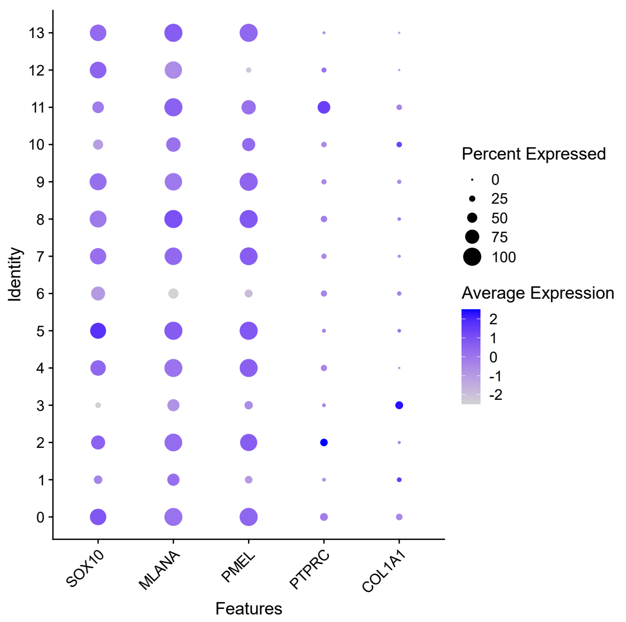
- Clusters that are close together on the UMAP share some top genes e.g., 1,3,4

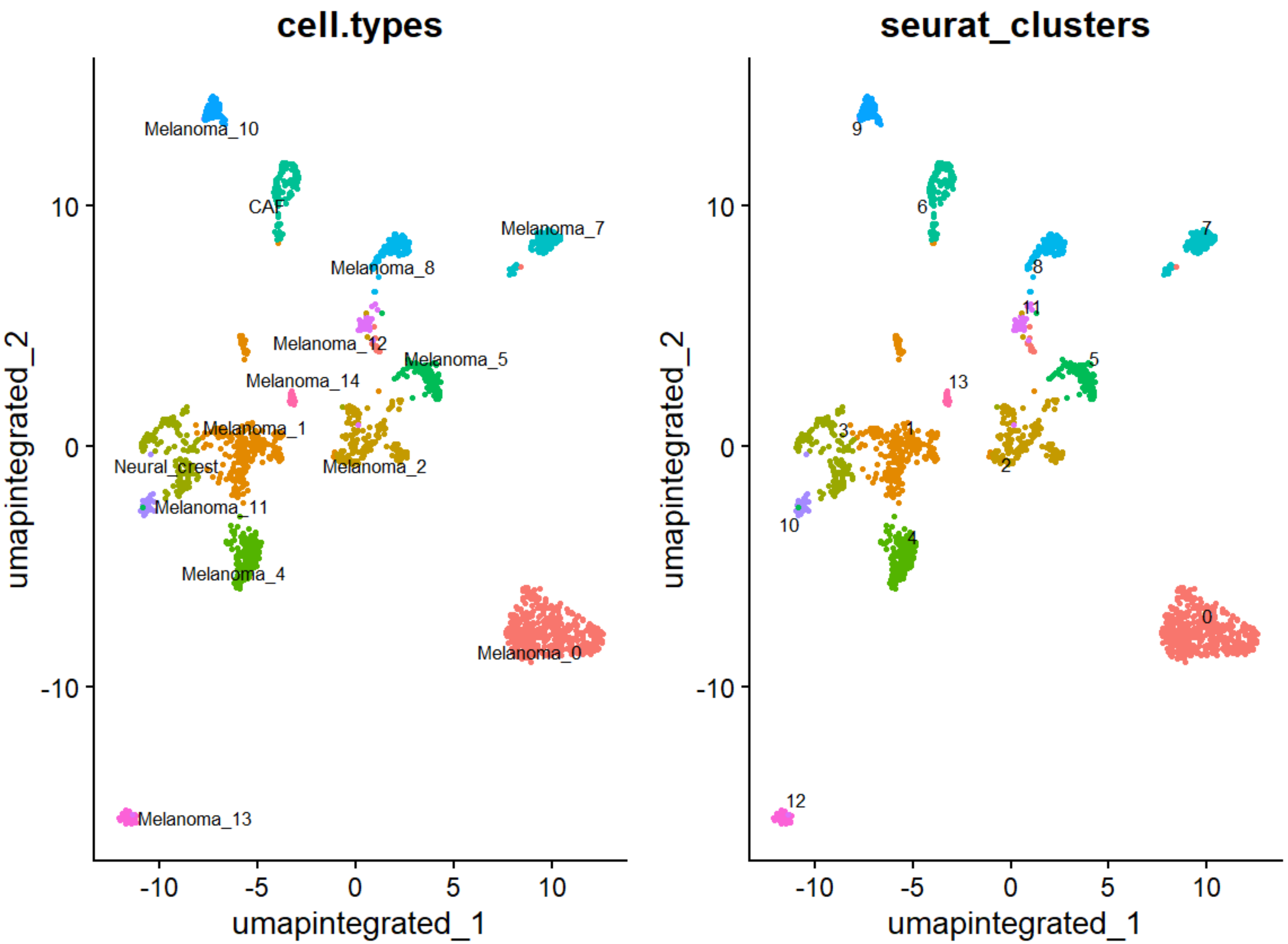


Cell Type Identification - Multipronged approach

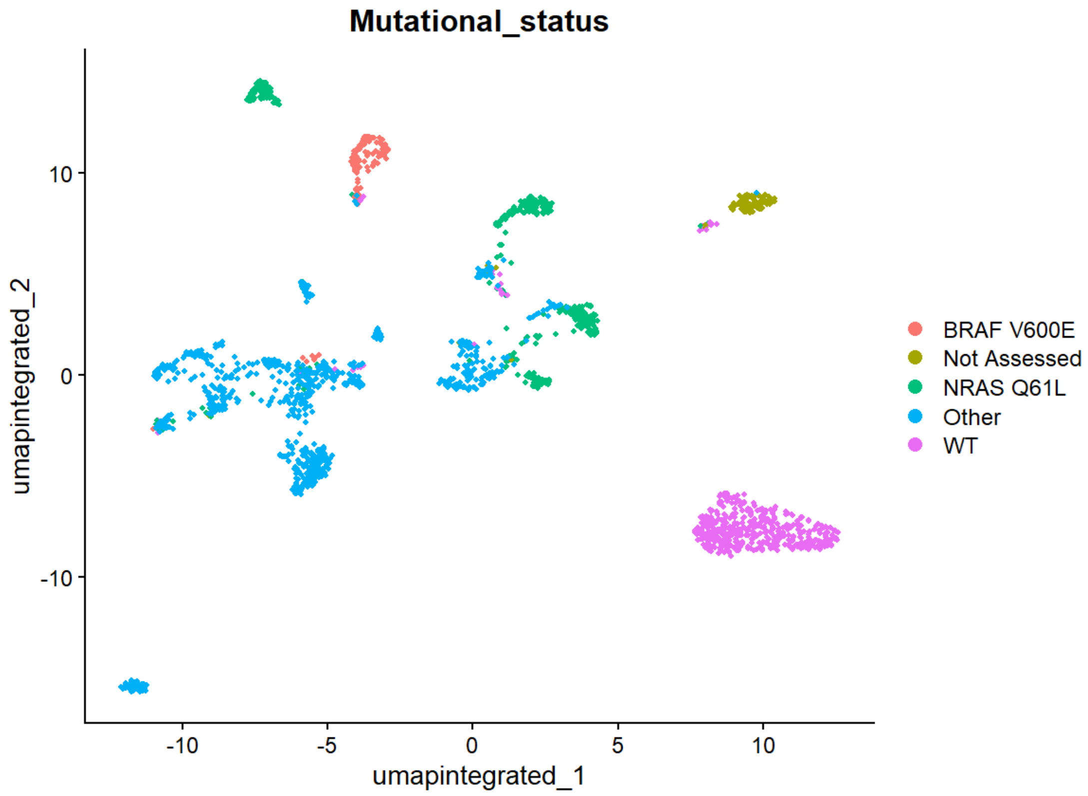
- Checked clusters for key melanoma cell markers
- Evaluated the percentage of these cell markers in the clusters
- Automated clustering – AZIMUTH & Lit review

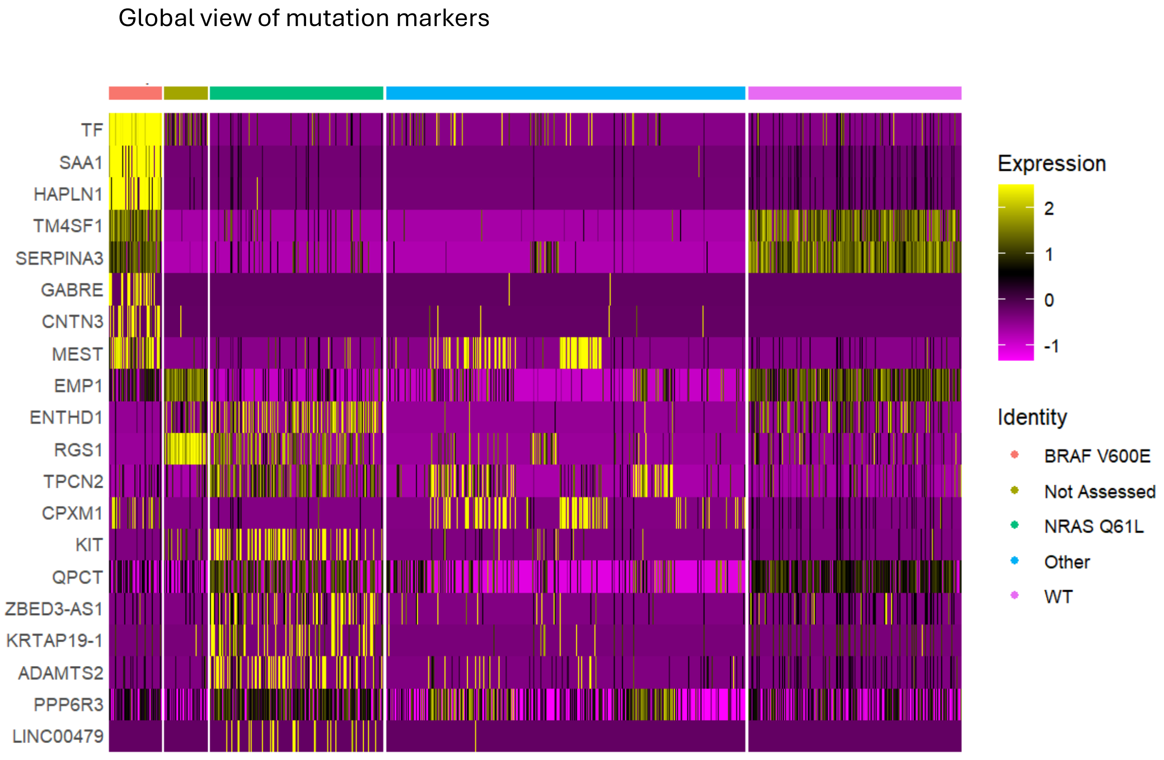




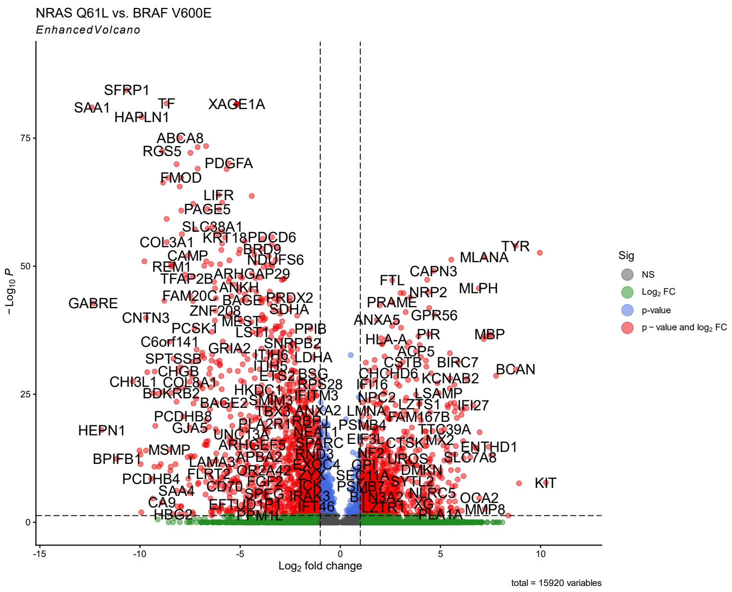


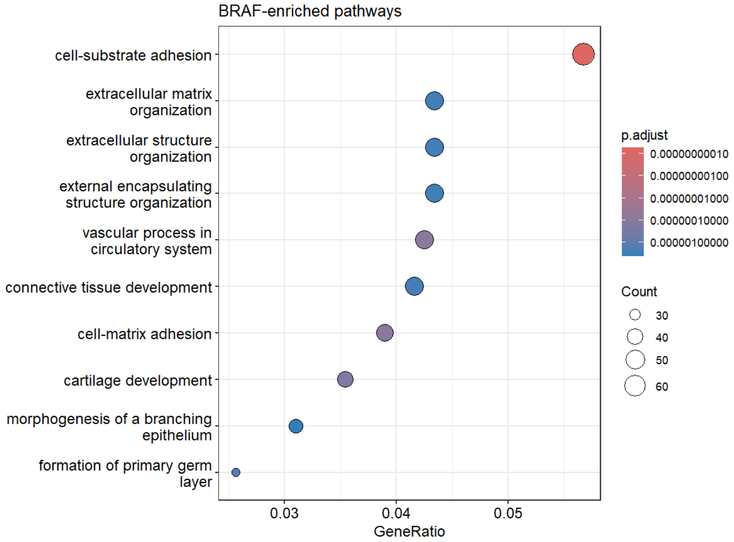
Hypothesis: Mutations like NRAS and BRAF mutations influence melanoma heterogeneity and cell signaling behavior





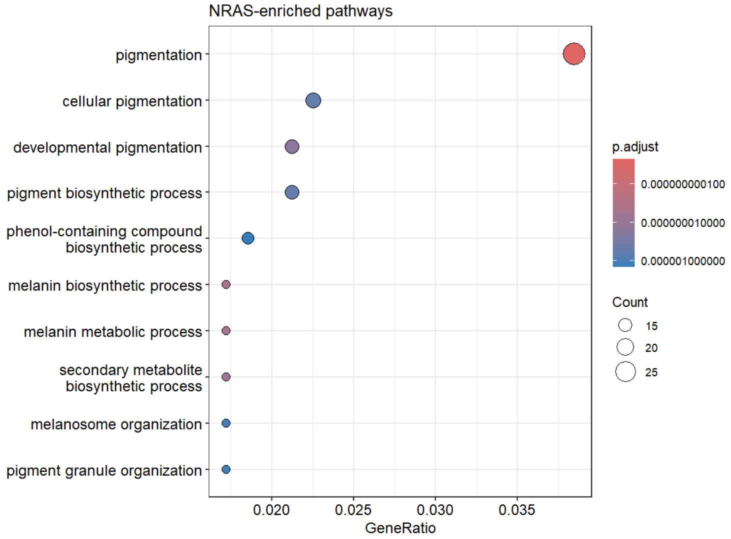
Distinct Gene Expression and Pathways in NRAS vs BRAF Melanoma





- Enriched for processes linked to:
  - Cell-substrate and matrix adhesion
  - Extracellular matrix organization
  - Tissue and structural development

✓ Suggests BRAF tumors **are less differentiated** and **more invasive**



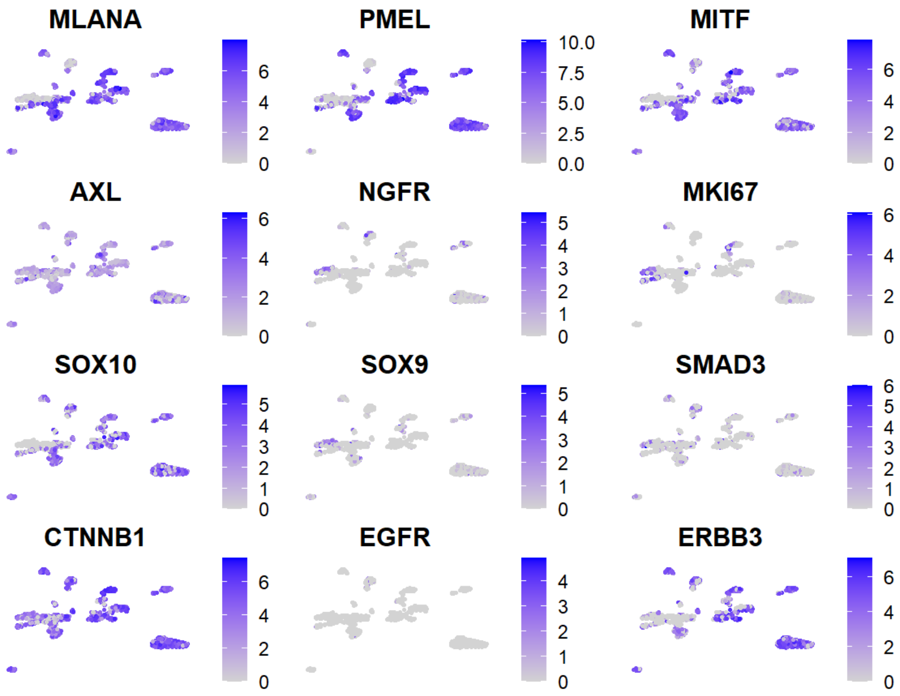
- Enriched for processes linked to:
  - Melanocyte differentiation
  - Pigmentation biology

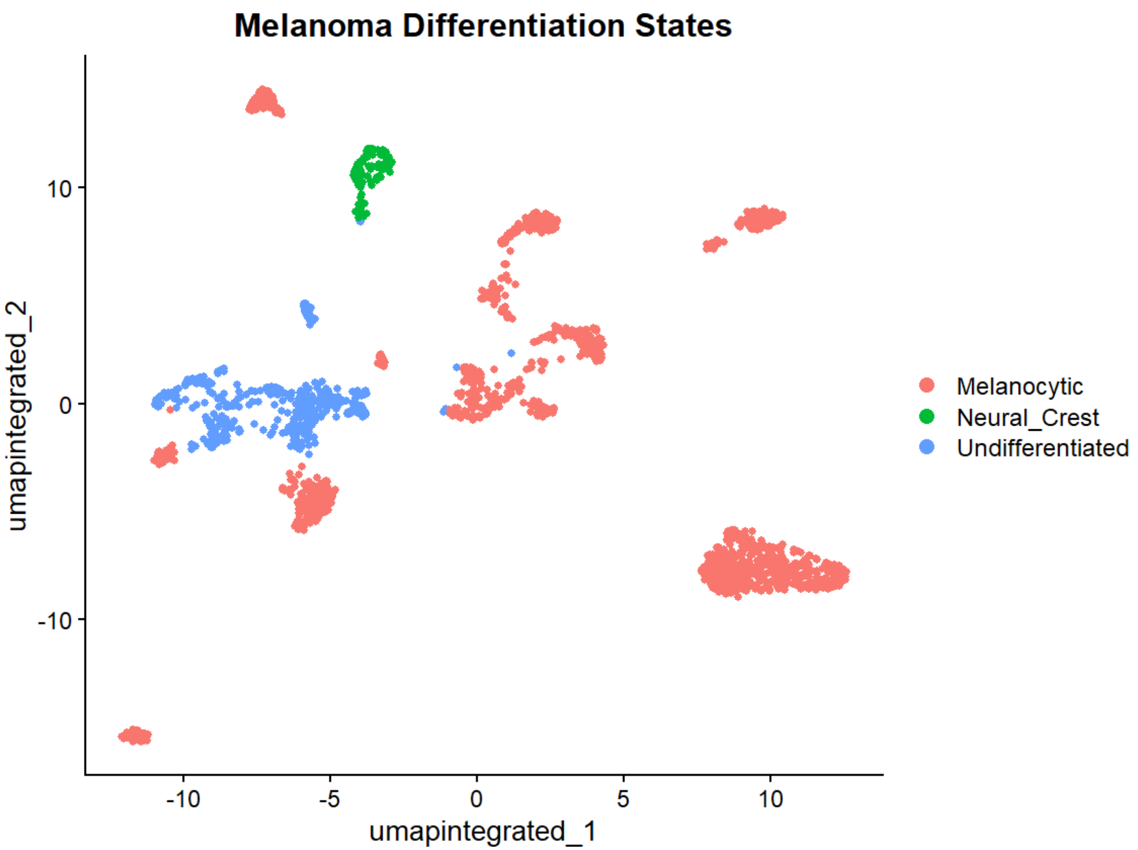
✓ Suggests NRAS tumours are **more differentiated**

Melanoma cells exist in four differentiation states, is this reflected in our cell clusters?

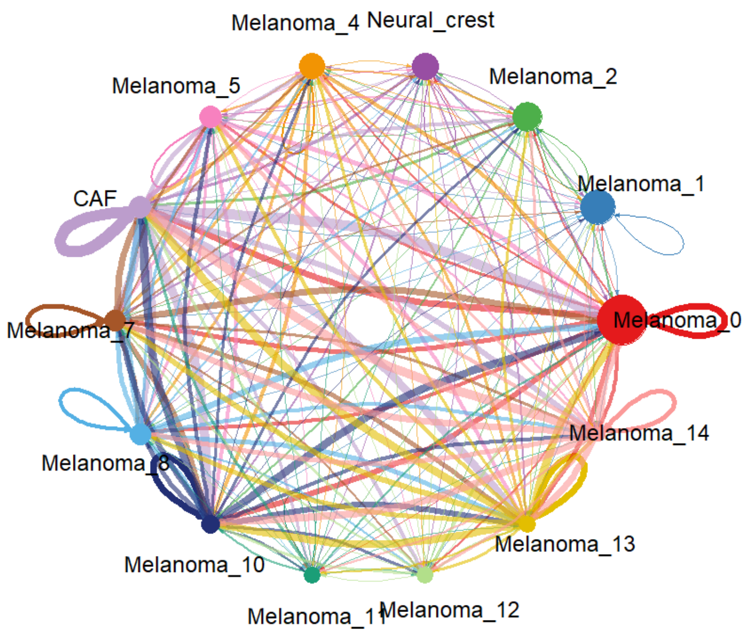
- Used marker genes + module scoring
- Assigned each cluster to a dominant differentiation state

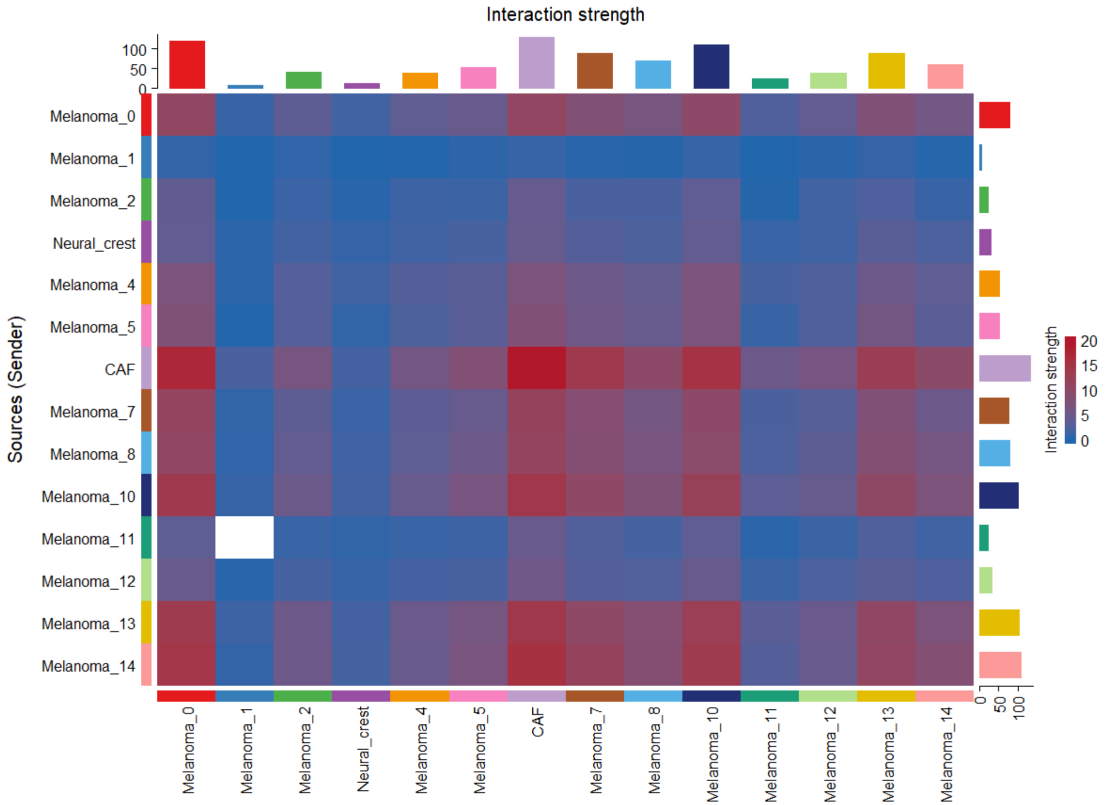
Differentiation State	Markers
Undifferentiated	AXL, SOX9, EGFR
Neural Crest	NGFR, SOX10, SMAD3
Transitory	MITF, SOX10, ERBB3
Melanocytic	MITF, CTNNB1, TYR, MLANA





Cell–Cell Communication in Melanoma



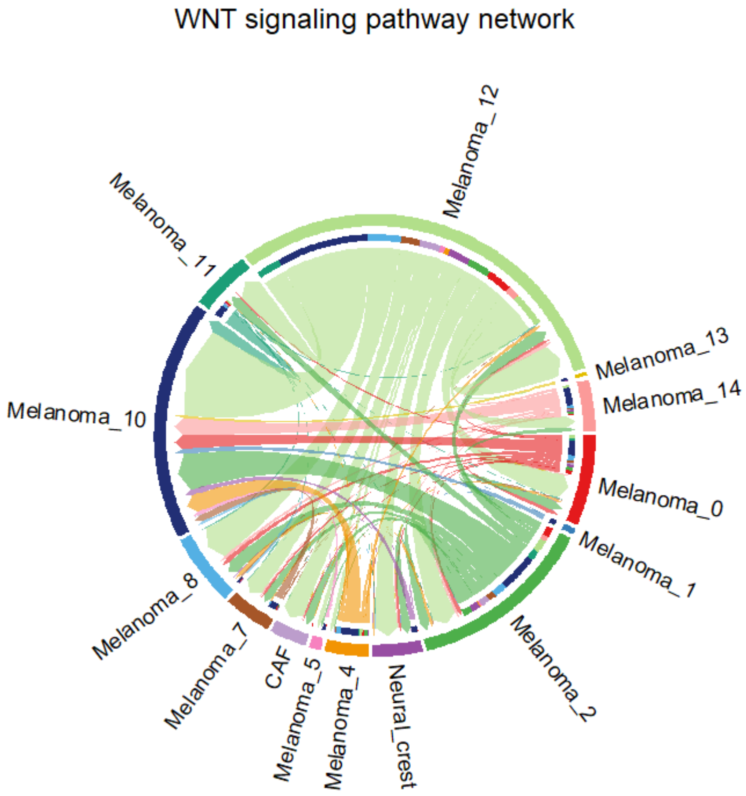


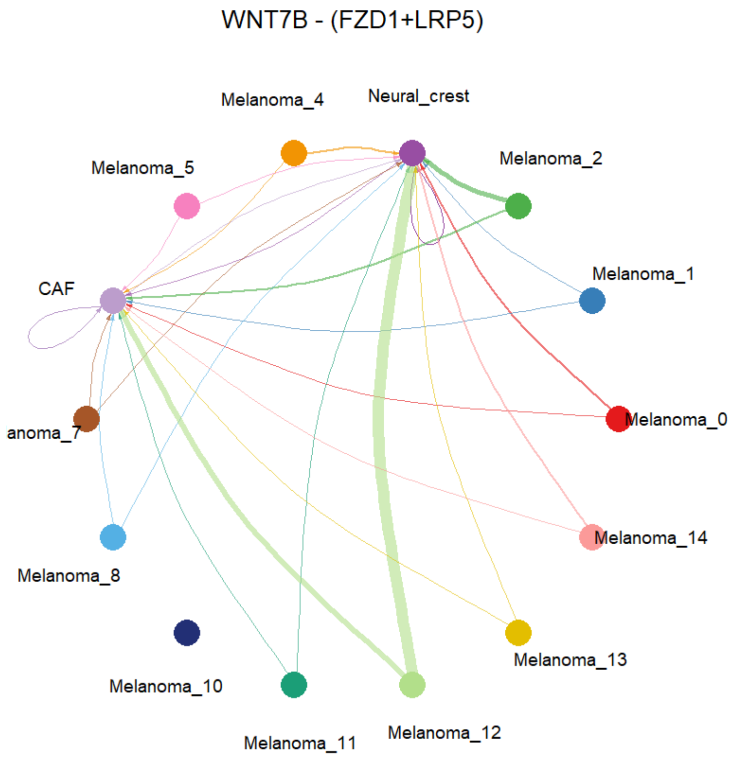
- Extensive communication across tumor and non-tumor cells
- Strong signaling interactions detected:
  - Between melanoma clusters and CAFs (Cancer-Associated Fibroblasts)
  - Within melanoma clusters (autocrine signaling)

Which signaling pathways drive these cell-cell interactions?

Focused on **the WNT signaling pathway**, known to play a key role in melanoma

- Extracted WNT-specific interactions from CellChat analysis
- Visualized interaction networks between cell types





- Identified key ligand-receptor pairs driving WNT signaling

Conclusion

- Melanoma shows high heterogeneity and multiple cell states (3)
- Mutations (NRAS vs BRAF) drive distinct tumor phenotypes
  - NRAS → more **differentiated**
  - BRAF → more **dedifferentiated / invasive**
- Tumor cells actively communicate with each other and CAFs
- WNT signaling underpins these interactions

Further Research

Investigate additional signaling pathways (e.g. **TGF-β, MAPK**) involved in tumor communication

Examine how different tumor states respond to **targeted therapies**

References

Acknowledgements

Dr. Alesandro Brombin

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